

Roll No.:

END SEMESTER EXAMINATION DECEMBER 2025

Name of the Program: B. Tech CSE Cyber Security

Semester: V

Name of the Course: Computer System Security

Course Code: TCS 597

Time: 3 Hours

Maximum Marks: 100

Note:

- i. All questions are compulsory.
- ii. Answer any two sub questions among a, b & c in each main questions.
- iii. Total marks in each main question are twenty.
- iv. Each questions carries 20 marks.

Q1	(20Marks)	
(a)	Explain the different types of SQL Injection attacks, i.e., in-band, inferential (blind), and out-of-band with examples.	CO1
(b)	What is a side-channel attack? Demonstrate its working mechanism using a figure.	CO2
(c)	Describe the Spectre attack. Explain its mechanism and why it is harder to mitigate than Meltdown.	CO5
Q2	(20Marks)	
(a)	Explain control hijacking attacks. How does buffer overflow attack contribute to such attacks? Use a programming example in your answer.	CO4
(b)	Explain different tools and techniques, which can be utilized for writing robust and secure application software.	CO2
(c)	What is a race condition vulnerability? Explain its causes, exploitation procedure, and mitigation strategies.	CO6
Q3	(20Marks)	
(a)	What is a format string vulnerability? Explain with an example how it can be exploited and list common ways to prevent it.	CO4
(b)	Demonstrate the working procedure of the CSRF attack. How this attack can be mitigated?	CO5
(c)	Explain the uses of the SSL protocol. Describe the steps involved in the SSL handshake process with a neat diagram.	CO2
Q4	(20Marks)	
(a)	Explain the concept of whole-disk encryption and its role in the smartphone data protection.	CO2
(b)	Discuss the access control mechanisms available for the Android operating system. How do they protect against unauthorized access attempts?	CO6

(c)	Explain static program analysis for vulnerability detection. Describe the common techniques used to identify vulnerabilities without executing the program.	CO3
Q5	(20Marks)	
(a)	Describe the process of multi-factor authentication (MFA). How does MFA enhance security compared to single-factor authentication?	CO5
(b)	Explain the main types of access control techniques, i.e., Discretionary Access Control (DAC), Mandatory Access Control (MAC), and Role-Based Access Control (RBAC). Compare their features and applications.	CO3
(c)	How malware can exploit zero-day vulnerabilities. Why are such attacks being more dangerous and difficult to mitigate?	CO2